Lab 9: Design and FPGA Implementation of the Control Path for a Single-Cycle RISC-V Processor

Objective

The goal of this laboratory exercise is to design and implement the control path of a single-cycle RISC-V (RV32I) processor using Verilog HDL. The design will be verified through simulation and synthesized on an FPGA for hardware demonstration. The control unit must support the following core RISC-V instructions:

ADD, ADDI, SUB, SLL, SRL, AND, OR, XOR, LW, LH, LB, SW, SH, SB, BEQ

Additionally, students will verify the implemented design using an FSM-based hardware verification approach.

Task 1: Developing the Control Path Truth Table

Objective

To determine the relationship between RISC-V instruction fields and the control signals required in a single-cycle datapath, expressed in the form of a truth table.

Procedure

1. Review the RISC-V Instruction Set Format

- Study the R-type, I-type, S-type, and B-type instruction formats.
- Identify the relevant fields: `opcode`, `funct3`, and `funct7`.

2. List Supported Instructions

The design must support the following RV32I instructions:

ADD, ADDI, SUB, SLL, SRL, AND, OR, XOR, LW, LH, LB, SW, SH, SB, BEQ

3. Identify Required Control Signals

Determine the control outputs needed for the datapath:

RegWrite, ALUSrc, MemRead, MemWrite, MemtoReg, Branch, ALUOp

4. Construct the Truth Tables

For each instruction, determine the correct setting of control signals based on the required datapath behavior. Use logic values:

$$1,\ 0,\ X \text{ (don't care)}$$

Deliverables

- A truth table for the main control unit (opcode as input, control signals as outputs).
- A truth table for the ALU control unit (ALUOp, funct3, funct7 as inputs).

Task 2: Verilog Description and Testbench Development

Objective

To implement the main control unit and ALU control logic in Verilog and verify their functionality through simulation using a testbench. The implementation must follow the truth tables created in Task 1.

Procedure

1. Implement the Main Control Unit

- **Inputs:** `opcode[6:0]`
- **Outputs:** RegWrite, ALUOp, MemRead, MemWrite, ALUSrc, MemtoReg, Branch

2. Implement the ALU Control Unit

- **Inputs:** ALUOp[1:0], `funct3`, `funct7`
- **Output:** ALUControl

3. Handle Don't-Care Conditions

Ensure that "don't care" (X) values from the truth table are assigned default or safe values in Verilog.

4. Develop the Testbench

The testbench must verify *every* supported instruction and generate waveforms for inspection.

5. Run Simulation

Simulate the design, generate waveforms, and verify correctness of the control signal transitions.

Deliverables

(a) Verilog source files for the Main Control and ALU Control modules.

(b) Testbench Verilog files.

(c) Simulation waveform screenshots highlighting all control signal transitions.

Task 3: FPGA Synthesis and RTL Analysis

Objective

To synthesize the RISC-V control path on an FPGA, examine the resulting RTL schematic, and verify functionality using physical switches and LEDs.

Procedure

(a) Integrate the MainControl and ALUControl modules into a top-level design.

(b) Implement a simple finite-state machine (FSM) inside the top-level module for verification.

(c) Connect input fields (`opcode`, `funct3`, `funct7`) to FPGA switches.

(d) Connect output control signals (RegWrite, MemRead, MemWrite, ALUSrc, MemtoReg, Branch, ALUControl, etc.) to LEDs.

(e) Use the FSM to read switch positions and update LED displays accordingly.

(f) Observe LED patterns to confirm that correct control signals are generated for each instruction.

(g) Synthesize the design and inspect the RTL schematic.

(h) Program the FPGA and verify correct operation by toggling switch inputs.

Deliverables

- RTL schematic screenshots.
- Demonstration of FPGA setup to the RA for approval.

Note: The switch interface, button debouncing module, and LED interface must be re used from Lab 5.

Assessment Rubric
Lab 9: Design and FPGA Implementation of the Control Path for a Single-Cycle RISC-V Processor

Name:	Student ID:	Section:

Points Distribution

	Task No.	LR 2 (Code)	LR 5 (Results)	LR 10 (Analysis)
In-Lab	Task 1	-	-	/20
	Task 2	/20	/15	-
	Task 3	/10	/15	-
Total Points		**/30**	**/30**	**/20**
CLO Mapped		CLO 1		

Affective Domain Rubric		Points	CLO Mapped
AR7	**Report Submission & Git Upload**	/10 & /10	CLO 1

CLO	Total Points	Points Obtained
1	100	
Total	**100**	

For description of different levels of the mapped rubrics, please refer to the Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics provided here.